SYSC 5108 Exam Study
W25 Sample Final Step-by-Step Solutions

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Source Note

This study note is based on the two files in the course directory:

- `W25_SYSC5108_Final.pdf`
- `W25_SYSC5108_Final_Solution.pdf`

The PDF filename says "Final", while the cover page itself says "Midterm I". I follow the file naming used in the directory and turn the provided sample exam and sample solution into a cleaner step-by-step study document.

Question 1: Linear Regression

Training points:

$$(0,-1),\ (1,2),\ (6,4),\ (9,2)$$

Test point:

$$(3,1).$$

We fit the line

$$\hat{y} = \beta_0 + \beta_1 x.$$

Step 1: Build the design matrix

$$X = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 6 \\ 1 & 9 \end{bmatrix}, \qquad y = \begin{bmatrix} -1 \\ 2 \\ 4 \\ 2 \end{bmatrix}.$$

Step 2: Use the normal equation

$$\beta = (X^\top X)^{-1} X^\top y.$$

Compute

$$X^\top X = \begin{bmatrix} 4 & 16 \\ 16 & 118 \end{bmatrix}, \qquad X^\top y = \begin{bmatrix} 7 \\ 44 \end{bmatrix}.$$

The determinant is

$$4(118) - 16(16) = 216.$$

So

$$(X^\top X)^{-1} = \frac{1}{216} \begin{bmatrix} 118 & -16 \\ -16 & 4 \end{bmatrix}.$$

Then

$$\beta = \frac{1}{216}\begin{bmatrix}118 & -16\\ -16 & 4\end{bmatrix}\begin{bmatrix}7\\ 44\end{bmatrix} = \begin{bmatrix}61/108\\ 8/27\end{bmatrix} \approx \begin{bmatrix}0.5648\\ 0.2963\end{bmatrix}.$$

Therefore the fitted model is

$$\boxed{\hat{y} = 0.5648 + 0.2963x.}$$

Step 3: Predict the training outputs

$$\hat{y}(0) = 0.5648, \quad \hat{y}(1) = 0.8611, \quad \hat{y}(6) = 2.3426, \quad \hat{y}(9) = 3.2315.$$

Step 4: Compute the training MSE

$$\mathrm{MSE}_{\mathrm{train}} = \frac{1}{4}\sum_{i=1}^{4}(y_i - \hat{y}_i)^2.$$

The squared errors are approximately

$$2.4481, \quad 1.2971, \quad 2.7467, \quad 1.5165.$$

So

$$\mathrm{MSE}_{\mathrm{train}} \approx \frac{2.4481 + 1.2971 + 2.7467 + 1.5165}{4} \approx 2.0021.$$

Rounded for exam work:

$$\boxed{\mathrm{MSE}_{\mathrm{train}} \approx 2.0.}$$

Step 5: Compute the test error

For the test input $x = 3$,

$$\hat{y}(3) = 0.5648 + 0.2963(3) \approx 1.4537.$$

The squared test error is

$$(1 - 1.4537)^2 \approx 0.2058.$$

Since there is only one test point, its MSE is the same number:

$$\boxed{\mathrm{MSE}_{\mathrm{test}} \approx 0.206.}$$

Question 2: Multiclass Logistic Regression Derivative

The exam gives

$$p(y = k \mid x, \beta) = \mathrm{softmax}(z)_k = \frac{e^{z_k}}{\sum_j e^{z_j}},$$

with

$$z_k = \beta_{k0} + \beta_{k1}x,$$

and

$$J(\beta) = -\sum_{i=1}^{M}\sum_{k=1}^{K} I_k(y^{(i)}) \log \mathrm{softmax}(z^{(i)})_k.$$

The target identity in the sample solution is

$$\nabla_{\beta_k} J(\beta_k) = -\sum_{i=1}^{M}\sum_{k=1}^{K} I_k(y^{(i)}) \left(1 - \mathrm{softmax}(z^{(i)})_k\right) x^{(i)}.$$

Step 1: Move the derivative inside the sum

$$\nabla_{\beta_k} J(\beta_k) = -\sum_{i=1}^{M}\sum_{k=1}^{K} I_k(y^{(i)}) \nabla_{\beta_k} \log \mathrm{softmax}(z^{(i)})_k.$$

Step 2: Differentiate the log

Use

$$\nabla_{\beta_k} \log \mathrm{softmax}(z)_k = \frac{\nabla_{\beta_k} \mathrm{softmax}(z)_k}{\mathrm{softmax}(z)_k}.$$

Step 3: Differentiate the softmax term

$$\mathrm{softmax}(z)_k = \frac{e^{z_k}}{\sum_j e^{z_j}}.$$

Using the quotient rule and the fact that

$$\nabla_{\beta_k} z_k = x,$$

we get

$$\nabla_{\beta_k} \mathrm{softmax}(z)_k = x \, \frac{e^{z_k}\left(\sum_j e^{z_j}\right) - e^{z_k} e^{z_k}}{\left(\sum_j e^{z_j}\right)^2}.$$

Factor out the softmax:

$$\nabla_{\beta_k} \mathrm{softmax}(z)_k = x \, \mathrm{softmax}(z)_k \left(1 - \mathrm{softmax}(z)_k\right).$$

Step 4: Divide by softmax$(z)_k$

$$\nabla_{\beta_k} \log \mathrm{softmax}(z)_k = x \left(1 - \mathrm{softmax}(z)_k\right).$$

Step 5: Substitute back into the loss gradient

$$\nabla_{\beta_k} J(\beta_k) = -\sum_{i=1}^{M}\sum_{k=1}^{K} I_k(y^{(i)}) x^{(i)} \left(1 - \mathrm{softmax}(z^{(i)})_k\right).$$

So the exam's target expression is

$$\boxed{\nabla_{\beta_k} J(\beta_k) = -\sum_{i=1}^{M}\sum_{k=1}^{K} I_k(y^{(i)}) \left(1 - \mathrm{softmax}(z^{(i)})_k\right) x^{(i)}}.$$

Study note

The sample solution differentiates the k-th softmax term with respect to the matching class parameter β_k. In standard multiclass logistic regression, the full familiar gradient is usually written as

$$\sum_{i=1}^{M} \left(p_k^{(i)} - I_k(y^{(i)})\right) x^{(i)}.$$

For exam study, it is useful to recognize both forms.

Question 3: 1D CNN Backpropagation

From the figure:

- input $= [1, 1, 1, 0, 0, 0]$,
- convolution bias $w_0 = 0.75$,
- shared kernel weights $w_1 = +1,\ w_2 = +1,\ w_3 = -1$,
- output-layer weights $w_{70} = 0.1,\ w_{75} = 0.2,\ w_{76} = 0.5$,
- ReLU activations everywhere,
- target output $t = 2$.

Step 1: Forward pass through the convolution layer

Window 1 uses $(1, 1, 1)$:

$$z_1 = 0.75 + 1(1) + 1(1) - 1(1) = 1.75, \qquad a_1 = \mathrm{ReLU}(z_1) = 1.75.$$

Window 2 uses $(1, 1, 0)$:

$$z_2 = 0.75 + 1(1) + 1(1) - 1(0) = 2.75, \qquad a_2 = 2.75.$$

Window 3 uses $(1, 0, 0)$:

$$z_3 = 0.75 + 1(1) + 1(0) - 1(0) = 1.75, \qquad a_3 = 1.75.$$

Window 4 uses $(0, 0, 0)$:

$$z_4 = 0.75, \qquad a_4 = 0.75.$$

Step 2: Max pooling

$$a_5 = \max(a_1, a_2) = \max(1.75, 2.75) = 2.75,$$

$$a_6 = \max(a_3, a_4) = \max(1.75, 0.75) = 1.75.$$

So the winning conv units are neuron 2 for pooling unit 5 and neuron 3 for pooling unit 6.

Step 3: Output neuron

$$z_7 = w_{70} + w_{75}a_5 + w_{76}a_6 = 0.1 + 0.2(2.75) + 0.5(1.75).$$

$$z_7 = 0.1 + 0.55 + 0.875 = 1.525.$$

Since ReLU is active,

$$a_7 = 1.525.$$

Part (a): Find the δ values

Using squared error $L = \frac{1}{2}(t - a_7)^2$,

$$\delta_7 = \frac{\partial L}{\partial z_7} = a_7 - t = 1.525 - 2 = -0.475.$$

Backpropagate to the pooling units:

$$\delta_5 = \delta_7 w_{75} = -0.475(0.2) = -0.095,$$

$$\delta_6 = \delta_7 w_{76} = -0.475(0.5) = -0.2375.$$

Because max pooling routes the gradient only to the winning input:

$$\delta_1 = 0, \qquad \delta_2 = -0.095, \qquad \delta_3 = -0.2375, \qquad \delta_4 = 0.$$

Therefore

$$\boxed{\delta_1 = 0,\ \delta_2 = -0.095,\ \delta_3 = -0.2375,\ \delta_4 = 0,\ \delta_5 = -0.095,\ \delta_6 = -0.2375,\ \delta_7 = -0.475}.$$

Part (b): Weight updates for w_{75} and w_1

For the fully connected weight:

$$\frac{\partial L}{\partial w_{75}} = \delta_7 a_5 = -0.475(2.75) = -1.30625.$$

So

$$\boxed{\frac{\partial L}{\partial w_{75}} = -1.30625 \approx -1.31.}$$

For the shared convolution weight w_1, add the contributions from each window:

$$\frac{\partial L}{\partial w_1} = \delta_1 x_1 + \delta_2 x_2 + \delta_3 x_3 + \delta_4 x_4.$$

The corresponding first inputs in the four windows are $1, 1, 1, 0$, so

$$\frac{\partial L}{\partial w_1} = 0(1) + (-0.095)(1) + (-0.2375)(1) + 0(0) = -0.3325.$$

Thus

$$\boxed{\frac{\partial L}{\partial w_1} = -0.3325.}$$

Question 4: Discounted Return

Given $\gamma = 0.9$ and reward sequence

$$R_1 = 2, \qquad R_2 = R_3 = \cdots = 7,$$

find G_0 and G_1.

Step 1: Compute G_1

$$G_1 = R_2 + \gamma R_3 + \gamma^2 R_4 + \cdots = 7 + 0.9(7) + 0.9^2(7) + \cdots .$$

This is a geometric series:

$$G_1 = 7\sum_{n=0}^{\infty} 0.9^n = \frac{7}{1-0.9} = 70.$$

Step 2: Compute G_0

$$G_0 = R_1 + \gamma G_1 = 2 + 0.9(70) = 2 + 63 = 65.$$

So

$$\boxed{G_1 = 70, \qquad G_0 = 65.}$$

Question 5: Reinforcement Learning

The environment is

State	Action	Next state	Probability	Reward
0	0	0	1	-1
0	1	1	1	1
1	0	1	1	1
1	1	0	1	-1

$$\gamma = 0.1, \qquad \pi(1 \mid 0) = \pi(0 \mid 1) = 0.6, \qquad \alpha = 0.6.$$

Part (a): Generate the trajectories

The random numbers are

$$0.94,\ 0.27,\ 0.30,\ 0.18,\ 0.75,\ 0.35.$$

Following the same action-selection convention as the sample solution:

- in state 0, choose action 1 with probability 0.6,
- in state 1, choose action 0 with probability 0.6.

Trajectory from state 0

- $0.94 > 0.4$, so choose action 1: $0 \to 1$, reward 1.
- $0.27 < 0.6$, so in state 1 choose action 0: $1 \to 1$, reward 1.
- $0.30 < 0.6$, so again choose action 0: $1 \to 1$, reward 1.

Thus

$$\tau_0 : 0 \to 1 \to 1 \to 1, \qquad r = (1, 1, 1).$$

Trajectory from state 1

- $0.18 < 0.6$, so choose action 0: $1 \to 1$, reward 1.
- $0.75 > 0.6$, so choose action 1: $1 \to 0$, reward -1.
- $0.35 < 0.4$, so in state 0 choose action 0: $0 \to 0$, reward -1.

Thus

$$\tau_1 : 1 \to 1 \to 0 \to 0, \qquad r = (1, -1, -1).$$

Monte Carlo values following the provided sample solution

The sample solution updates to the following values:

$$v(0) \approx -0.156, \qquad v(1) \approx 0.974.$$

For the action values:

$$q(0,0) = -1, \qquad q(0,1) = 1 + 0.1(1) + 0.1^2(1) = 1.11,$$

$$q(1,0) \approx 0.974, \qquad q(1,1) = -1 + 0.1(-1) = -1.1.$$

So the sample-final answer is

$$\boxed{v(0) \approx -0.156, \quad v(1) \approx 0.974}$$

and

$$\boxed{q(0,0) = -1, \quad q(0,1) = 1.11, \quad q(1,0) \approx 0.974, \quad q(1,1) = -1.1}.$$

Part (b): Off-policy TD control (Q-learning)

Use

$$q(s_t, a_t) \leftarrow q(s_t, a_t) + \alpha \left[r_{t+1} + \gamma \max_{a'} q(s_{t+1}, a') - q(s_t, a_t) \right].$$

Start from

$$q(s,a) = 0 \quad \text{for all } (s,a).$$

Update along τ_0

$$q(0,1) = 0 + 0.6[1 + 0.1(0) - 0] = 0.6.$$

$$q(1,0) = 0 + 0.6[1 + 0.1(0) - 0] = 0.6.$$

$$q(1,0) = 0.6 + 0.6[1 + 0.1(0.6) - 0.6] = 0.6 + 0.6(0.46) = 0.876.$$

Update along τ_1

$$q(1,0) = 0.876 + 0.6[1 + 0.1(0.876) - 0.876] \approx 1.003.$$

$$q(1,1) = 0 + 0.6[-1 + 0.1(0.6) - 0] = -0.564.$$

$$q(0,0) = 0 + 0.6[-1 + 0.1(0.6) - 0] = -0.564.$$

Therefore

$$\boxed{q(0,0) = -0.564, \quad q(0,1) = 0.6, \quad q(1,0) \approx 1.003, \quad q(1,1) = -0.564}.$$

Question 6: Bayesian Network

Variables:

- S: storm,
- B: burglar,
- C: cat,
- A: alarm.

Part (a): Directed model

The natural causal graph is

$$S \to B, \qquad S \to C, \qquad B \to A, \qquad C \to A.$$

Reason:

- storms affect whether burglars show up,
- storms affect whether cats come inside,
- burglars and cats both affect whether the alarm is triggered.

Part (b): Estimate the probabilities

The data table has 13 samples.

1. Prior for storm

Storm is true in IDs $10, 11, 12, 13$, so

$$P(S = T) = \frac{4}{13} \approx 0.3077.$$

2. Burglar given storm

When $S = T$, only ID 13 has $B = T$, so

$$P(B = T \mid S = T) = \frac{1}{4} = 0.25.$$

When $S = F$, IDs $7, 8, 9$ have $B = T$, so

$$P(B = T \mid S = F) = \frac{3}{9} = \frac{1}{3} \approx 0.3333.$$

3. Cat given storm

When $S = T$, IDs $10, 11, 12$ have $C = T$, so

$$P(C = T \mid S = T) = \frac{3}{4} = 0.75.$$

When $S = F$, IDs $6, 9$ have $C = T$, so

$$P(C = T \mid S = F) = \frac{2}{9} \approx 0.2222.$$

4. Alarm given burglar and cat

$$P(A = T \mid B = T, C = T) = \frac{1}{1} = 1.0$$

because ID 9 is the only case.

$$P(A = T \mid B = T, C = F) = \frac{2}{3} \approx 0.6667$$

from IDs $7, 8, 13$, where alarm is true in $8, 13$.

$$P(A = T \mid B = F, C = T) = \frac{1}{4} = 0.25$$

from IDs $6, 10, 11, 12$, where alarm is true only in 10.

$$P(A = T \mid B = F, C = F) = \frac{1}{5} = 0.2$$

from IDs $1, 2, 3, 4, 5$, where alarm is true only in 5.

So the CPTs are:

$$\boxed{P(S = T) = 0.3077}$$

$$\boxed{P(B = T \mid S = T) = 0.25, \quad P(B = T \mid S = F) = 0.3333}$$

$$\boxed{P(C = T \mid S = T) = 0.75, \quad P(C = T \mid S = F) = 0.2222}$$

$$\boxed{\begin{aligned} &P(A = T \mid B = T, C = T) = 1.0, \\ &P(A = T \mid B = T, C = F) = 0.6667, \\ &P(A = T \mid B = F, C = T) = 0.25, \\ &P(A = T \mid B = F, C = F) = 0.2. \end{aligned}}$$

Part (c): Predict ALARM when $B = T$, $C = T$, $S = F$

Once B and C are known, A depends only on B and C, not directly on S.

From the CPT:

$$P(A = T \mid B = T, C = T) = 1.$$

Therefore the Bayesian network predicts

$$\boxed{A = T.}$$

Part (d): Generate one sample

Use the random numbers

$$0.94, \ 0.27, \ 0.30, \ 0.18.$$

Following the same sampling convention as the sample solution, assign the "false" interval first and the "true" interval second.

1. Sample S

$$P(S = T) = 0.3077 \quad \Rightarrow \quad P(S = F) = 0.6923.$$

Since

$$0.94 > 0.6923,$$

we get

$$S = T.$$

2. Sample $B \mid S = T$

$$P(B = T \mid S = T) = 0.25 \quad \Rightarrow \quad P(B = F \mid S = T) = 0.75.$$

Since

$$0.27 < 0.75,$$

we get

$$B = F.$$

3. Sample $C \mid S = T$

$$P(C = T \mid S = T) = 0.75 \quad \Rightarrow \quad P(C = F \mid S = T) = 0.25.$$

Since

$$0.30 > 0.25,$$

we get

$$C = T.$$

4. Sample $A \mid B = F, C = T$

$$P(A = T \mid B = F, C = T) = 0.25 \quad \Rightarrow \quad P(A = F \mid B = F, C = T) = 0.75.$$

Since

$$0.18 < 0.75,$$

we get

$$A = F.$$

Therefore the generated sample is

$$\boxed{S = T, \quad B = F, \quad C = T, \quad A = F.}$$

Final Answers Summary

- Q1:

$$\hat{y} = 0.5648 + 0.2963x, \quad \text{MSE}_{\text{train}} \approx 2.0, \quad \text{MSE}_{\text{test}} \approx 0.206.$$

- Q2: use the provided softmax derivative chain to obtain

$$\nabla_{\beta_k} J(\beta_k) = -\sum_{i=1}^{M} \sum_{k=1}^{K} I_k(y^{(i)}) \left(1 - \text{softmax}(z^{(i)})_k\right) x^{(i)}.$$

- Q3:

$$\delta_1 = 0,\ \delta_2 = -0.095,\ \delta_3 = -0.2375,\ \delta_4 = 0,\ \delta_5 = -0.095,\ \delta_6 = -0.2375,\ \delta_7 = -0.475$$

and

$$\frac{\partial L}{\partial w_{75}} = -1.30625, \qquad \frac{\partial L}{\partial w_1} = -0.3325.$$

- Q4:

$$G_0 = 65, \qquad G_1 = 70.$$

- Q5:

$$v(0) \approx -0.156, \quad v(1) \approx 0.974$$

  and

$$q(0,0) = -1,\ q(0,1) = 1.11,\ q(1,0) \approx 0.974,\ q(1,1) = -1.1$$

  for the sample-solution Monte Carlo part, and

$$q(0,0) = -0.564,\ q(0,1) = 0.6,\ q(1,0) \approx 1.003,\ q(1,1) = -0.564$$

  for Q-learning.

- Q6:

$$P(S=T) = 0.3077, \quad P(B=T \mid S=T) = 0.25, \quad P(B=T \mid S=F) = 0.3333,$$
$$P(C=T \mid S=T) = 0.75, \quad P(C=T \mid S=F) = 0.2222,$$
$$P(A=T \mid T,T) = 1, \quad P(A=T \mid T,F) = 0.6667, \quad P(A=T \mid F,T) = 0.25, \quad P(A=T \mid F,F) = 0.2,$$

  with prediction $A = T$ for $B = T, C = T, S = F$, and generated sample

$$(S,B,C,A) = (T,F,T,F).$$